CYBERGUARD SENTINEL

AI-Based User Behavior Anomaly Detection Platform

REPORT CLASSIFICATION: CONFIDENTIAL
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1. Executive Security Summary

This report summarizes behavioral telemetry, machine learning anomaly detection scores, and security rule evaluations across monitored user accounts. Anomalies are detected via Scikit-Learn Isolation Forest paired with behavioral baseline profiling and rule heuristics.

Total Monitored Events	ML Anomaly Detections	High-Risk Incidents	Open Security Alerts
95	25	25	39

2. Risk Score & Distribution Analysis

Risk Tier	Score Range	Event Count	Proportion (%)	Operational Policy
LOW RISK	0 - 29	52	54.7%	Standard logging; automatic baseline continuous learning
MEDIUM RISK	30 - 69	18	18.9%	Elevated monitoring; baseline freeze; SOC analyst triage queue
HIGH RISK	70 - 100	25	26.3%	Immediate alert raised; session termination / MFA step-up candidate

3. High-Risk & Suspicious Activity Log (Recent 8 Incidents)

Timestamp (UTC)	User	IP / Location	Target Resource	Risk Score	Flagged Detection Details
2026-11-01 15:31	test.user	45.154.255.89 Live Account Ingress (45.154.255.89)	/api/v1/internal/admin-dump	**100.0** (High)	Isolation Forest identified multidimensional behavioral divergence (ML Outlier Score: 0.60).; Authentication originated from unrecognized IP address: 45.154.255.89.
2026-11-01 15:31	test.user	10.0.12.14 Live Account Ingress (10.0.12.14)	/dashboard	**79.6** (High)	Unfamiliar device hardware signature: Desktop Workstation (Corporate Dell).; Impossible Travel Detected: Ingress shifted 6,711 km from London to Delhi in 287.4 minutes (Implied Velocity: 1,401 km/h - Commercial Jet Limit: 800 km/h).
2026-11-01 15:19	vadmin	127.0.0.1 Internal Portal Ingress	/login	**34.8** (Medium)	Elevated credential failure velocity: 2 prior failed attempts.
2026-11-01 10:44	test.user	45.154.255.89 Live Account Ingress (45.154.255.89)	/api/v1/internal/admin-dump	**100.0** (High)	Isolation Forest identified multidimensional behavioral divergence (ML Outlier Score: 0.60).; Authentication originated from unrecognized IP address: 45.154.255.89.
2026-11-01 10:43	test.user	45.154.255.89 Live Account Ingress (45.154.255.89)	/api/v1/internal/admin-dump	**100.0** (High)	Isolation Forest identified multidimensional behavioral divergence (ML Outlier Score: 0.60).; Authentication originated from unrecognized IP address: 45.154.255.89.

2026-11-01 09:41	test.user	127.0.0.1 Internal Portal Ingress	/dashboard	**59.8** (Medium)	Critical brute-force attack threshold reached: 5 repeated authentication failures.
2026-11-01 09:41	test.user	127.0.0.1 Internal Portal Ingress	/dashboard	**34.3** (Medium)	Elevated credential failure velocity: 4 prior failed attempts.
2026-11-01 09:41	test.user	127.0.0.1 Internal Portal Ingress	/dashboard	**33.9** (Medium)	Elevated credential failure velocity: 3 prior failed attempts.

4. MITRE ATT&CK; Adversarial Tactics & Technique Alignment

Security events, behavioral deviations, and Isolation Forest ML anomalies are correlated directly with official MITRE ATT&CK; Enterprise Matrix v14 tactics and techniques to facilitate standardized SOC response:

Technique ID	Tactic Category	Adversary Technique Name	Detection Heuristic & Indicator
T1110.001	Credential Access	Brute Force: Password Guessing	Adversary repeatedly attempts authentication passwords against valid account names to gain unauthorized access.
T1078.004	Defense Evasion & Initial Access	Valid Accounts: Cloud / Enterprise Accounts	Adversary leverages stolen credentials from geographically disparate locations, violating Haversine physical velocity limits.
T1083	Discovery & Credential Access	File and Directory Discovery	Adversary enumerates and probes sensitive application assets including credential stores, system configuration, and payroll vaults.
T1588.002	Resource Development	Obtain Capabilities: Penetration Testing Tool	Adversary interacts with authentication endpoints via adversarial distributions (Kali Linux, Parrot OS, Tor Browser) or automated scraping utilities.
T1071.001	Command and Control	Application Layer Protocol: Web Protocols	Adversary establishes authentication session through previously unobserved foreign IP, proxy, or anonymous ASN infrastructure.
T1078	Defense Evasion	Valid Accounts: Schedule Anomaly	Authentication attempted during anomalous off-hours schedule (e.g. 02:00 AM) to circumvent active administrator vigilance.
T1204	Execution	User Execution: Rogue Client Device	Session initiated from an unrecognized hardware device signature or spoofed User-Agent environment.
T1078	Initial Access	Valid Accounts: Multi-Vector Outlier	Isolation Forest identified extreme multi-dimensional divergence across timing, velocity, frequency, and duration dimensions.

5. Recommended Security Actions & Remediation

1. Multi-Factor Authentication (MFA) Step-Up: Enforce mandatory FIDO2 / TOTP challenge for logins originating from new IP addresses or unfamiliar client operating systems.

2. Account Lockout & Throttling: Configure automated temporary lockout or exponential rate-limiting for accounts accumulating 3+ failed logins in 5 minutes.

3. Baseline Safeguard Enforcement: Keep behavioral baseline profiles locked for users undergoing internal investigation to prevent adversarial profile poisoning.

4. Geofencing & Off-Hours Ingress Policies: Implement conditional access policies restricting administrative asset access between 23:00 and 06:00 UTC without prior SOC clearance.

5. SOC Incident Response: Review all OPEN High and Critical security alerts within a 15-minute SLA. Resolve alerts with thorough forensic notes.